



Fast processing and classification of epileptic seizures based on compressed EEG signals

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ABSTRACT

The diagnosis of epilepsy based on visual inspection of electroencephalogram (EEG) signals is inherently complex and prone to error, even for physicians, mainly due to the large number of signals involved and the variability between individuals. These same challenges make the development of portable epilepsy diagnostic systems for everyday use difficult. Key obstacles include the immense complexity of signal processing and the inherent ambiguity in accurately classifying disease. For these reasons, we propose in this paper the deployment of compressive sensing to condense EEG signals while preserving relevant information, allowing seizure classification based on systematically selected features of the reconstructed signals. Based on a dataset comprising EEG recordings from 13 epileptic patients with various seizure types, we explore the deployment of the discrete cosine transform (DCT) and random matrix multiplication for compression ratios ranging from 5% to 70%, balancing data reduction with signal fidelity. Following the extraction of relevant features, selection was performed based on mutual information and a correlation matrix to preserve only the most relevant features for analysis. For classification, following a comparison of adequate machine learning models, XGBoost is chosen as it realizes a classification accuracy of 98.78%. The CS method was implemented on an STM32 microcontroller and a Raspberry Pi for reconstruction and classification, to demonstrate feasibility as an embedded system. At 70% compression, significant improvements have been observed: 70% file size reduction, 84% decrease in transmission time (from 2518.532s to 400.392s), and substantial energy savings (e.g., from 11.5±0.707 mWh to 4.5±0.707 mWh for Patient 12). Thereby, the signal quality was maintained with PSNR of 16.15±3.98 and Pearson correlation coefficient of 0.68±0.15. The proposed system highlights the potential for efficient, portable, real-time epilepsy diagnosis systems that achieve precise and fully automated seizure classification.

1. Introduction

Epilepsy, a chronic neurological disorder that causes recurrent and unprovoked seizures, affects people of all ages around the world. According to the World Health Organization (WHO), more than 50 million people suffer from epilepsy, which represents approximately 1% of the world's population [1]. Despite advances in neuroimaging and electroencephalography (EEG), the precise identification and classification of epileptic seizures remain complex tasks, often requiring expert interpretation of extensive EEG recordings. The need for more efficient, accurate, and accessible diagnostic tools is crucial, particularly in resource-limited settings where expert neurologists may be scarce.

To effectively diagnose epilepsy, it is important to obtain a detailed description of the seizure pattern. One of the most reliable and established methods for epilepsy diagnosis is electroencephalography (EEG) [2]. Accurate seizure classification is of great importance as it also affects medication choices. However, differentiating between various types of seizures is challenging due to the similarity of EEG signs and the presentation of disparate symptoms in individuals with the same type of seizures [3]. Examining EEGs for epileptic seizures has traditionally involved manual inspection by certified EEG interpreters. This process requires the observer to monitor the patient for

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an extended period, which can be challenging, particularly when the EEG signals have been recorded over an extended duration [4]. This process also places a significant physical and psychological burden on clinicians, as EEG recordings typically take several hours, and some patients must be monitored overnight or even for several days [5]. Moreover, relying on the manual review of extended EEG recordings introduces the possibility of human error [6].

Given these numerous challenges, a significant amount of research has been directed towards developing highly precise machine learning and deep learning models for seizure classification [7,8]. However, the automatic classification of the type of seizure has been relatively overlooked, due to two main factors: The inherent complexity of solving the classification problem of the type of seizure and the lack of available clinical data [9]. Due to the lack of publicly available EEG data and the complexity of clinical annotation, automating the process of differentiating between focal, and generalized EEG data remains an overlooked and technically challenging area.

Mozafari et al. [10], developed a technique to classify focal and generalized seizures. They used blind source separation to remove artifacts from EEG signals, divided channels into two groups, applied thresholding, and determined the seizure class based on activity in each group. This method produced 80.72% accuracy, 80% sensitivity, 81.08% specificity, and 67.55% precision in a mixed seizure data set. Einizade et al. [11], proposed using deep RNN and CNN to classify generalized and focal seizures, allowing the capture of spatial and temporal information. Their approach achieved 82% precision, 71.69% precision, and 85% sensitivity. George et al. [12], developed a seizure classification method based on the tunable Q wavelet transform (TQWT), entropy measures, particle swarm optimization (PSO), and artificial neural networks (ANN). EEG data were partitioned into subbands using TQWT and features such as log energy, Shannon entropy, and Stein's unbiased risk estimate (SURE) entropy were computed. These features were then used to identify informative features, which were then fed into an ANN classifier. Their method achieved 88.8% precision in classifying normal, focal, and generalized seizures.

To date, numerous studies have been conducted to realize automated seizure detection in embedded systems. A study by Ohannesian et al. [13], proposed a hybrid system of supervised and unsupervised machine learning algorithms for detecting epileptic seizures from EEG data. A discrete wavelet transform (DWT) was implemented to decompose the EEG signal into frequency subbands, and the Gaussian Mixture Model (GMM) was used to partition the extracted features. The proposed system aimed to increase detection accuracy while reducing execution time. Another study by Sarić et al. [14], proposed the use of a field-programmable gate array (FPGA)-based method to achieve classification and received the highest accuracy score to classify focal and generalized seizures. The proposed approach utilized a feedforward multilayer perceptron (MLP ANN) to extract five key features from the EEG signals, employing techniques like time–frequency analysis, continuous wavelet transform (CWT), and statistical analysis. Then, the appropriate parameters of the ANN model were found using k-fold cross-validation. The best-performing ANN model was used to perform real-time classification on this model, which was programmed in the very high-speed integrated circuit hardware description language (VHDL) on an FPGA. With the 5-12-3 MLP ANN set-up, the accuracy of this investigation was 95.14%.

Even if the classification results are satisfactory, the practical applicability of the proposed systems in real-time diagnostic of epileptic seizures remains hindered by several limitations [15]. Many challenges arise related to computational overhead, limited available resources concerning memory allocation or storage space, high dimensionality of EEG signals, huge energy and time consumption during EEG data transfer, and classification model performance [16]. In this paper, we follow the research path to classify epileptic seizures and propose a new method based on compressive sensing (CS) [17], which is a signal processing approach that efficiently compresses EEG signals by

exploiting signal sparsity to reduce the number of samples to even less than required by the Nyquist–Shannon sampling theorem. Reducing the amount of data before signal transmission significantly reduces transmission time [18]. We propose, therefore, to use compressive sensing and machine learning classification to pave the way to faster and energy-efficient epileptic seizure classification systems.

The paper consists of seven sections. Section 2 provides an in-depth look at the EEG dataset used in the study, outlining the preprocessing steps essential for ensuring data integrity and relevance. This includes details on data acquisition, noise reduction techniques, and artifact removal methods. Section 3 introduces compressive sensing techniques, covering the processes involved in signal compression, reconstruction, and the evaluation metrics used to assess the effectiveness of these operations. Section 4 discusses the extraction and selection of features from the EEG signals, as well as the methodologies employed to evaluate these features for their efficacy in seizure classification. In Section 5, the focus shifts to the development and evaluation of various machine learning models tailored for seizure classification, including their training, validation, and performance assessment. Section 6 provides a comparison of the proposed method with existing state-of-the-art techniques. Finally, Section 7 concludes the paper by summarizing the key findings, discussing their broader implications, and suggesting potential directions for future research.

2. Methodology

2.1. EEG dataset

The EEG data were obtained from the Pediatric Neurology Unit of Sfax, Tunisia. They were collected from 13 pediatric patients with intractable seizures (7 females, 2–22 years old, and 6 males, 8–14 years old) at Hedi Chaker Hospital (CHU) in Sfax to evaluate their potential for surgical intervention. The recordings are organized into 13 cases and were obtained at a sampling rate of 256 samples per second and a 16-bit resolution from electrodes placed according to the International 10-20 Electrode Positioning System. The total duration of the available EEG recordings is approximately 34 min. In 13 cases, EEG recordings, including the preictal state and ictal activity, for the onset of seizures and offset time intervals after visual inspection resulted in a total of 28 seizures. The identification of the types of seizures in the patient is documented in the clinical reports and their EEG scans. These reports contain unstructured text, including medical history, medications, and clinical evaluations. The annotation team classified different types of seizures based on these reports and preliminary EEG analysis. These seizures were then classified by the experts into three different types: Asymmetric tonic seizures (AsTSZ), symmetric tonic seizures (STSZ), and seizures of unknown origin (USZ).

Symmetrical tonic seizures are characterized by simultaneous symmetric contractions of the muscles in the arms and legs, often causing the individual to fall or have difficulty moving [19]. On the other hand, asymmetrical tonic seizures (AsTSZ) are characterized by asymmetrical tonic posturing of the extremities, meaning that posturing is not equal on both sides of the body. These seizures usually originate in the supplementary motor area and are usually a manifestation of frontal lobe epilepsy. Asymmetric tonic posturing is more commonly observed in the upper extremities contralateral to the epileptogenic cortex [20]. It may also affect the lower extremities and the head. In some cases, the seizure may be preceded by sensations such as a sensation of lightning in the body. The unknown type of seizure indicates that a patient is experiencing some type of seizure. However, it is difficult to accurately determine the appropriate type due to insufficient and complex EEG information [14]. Additional information about the EEG dataset used in this study is presented in Table 1. As can be seen from the annotation files in the database, in most of the 13 cases, there are frequent changes in the recording assembly of the EEG signal, with channels being added or removed from one epoch to another during the

Table 1
Recorded EEG dataset.

Subject ID	Gender	Age (years)	Recording time (s)	Number of seizures	Seizure type
1	M	6	36	1	STSZ
2	M	10	93	1	STSZ
3	F	7	39	1	STSZ
4	F	9	127	2	STSZ
5	F	1	553	5	STSZ
6	F	17	44	1	AsTSZ
7	F	9	131	2	AsTSZ
8	M	10	80	4	AsTSZ
9	F	22	221	5	AsTSZ
10	M	10	86	2	USZ
11	F	11	30	1	USZ
12	F	5	580	2	USZ
13	M	7	37	1	USZ
Total:	5M/7F	average = 9.5	33 min and 57 s	28	3 classes

recording process. Since all EEG epochs are used interchangeably for training and testing through cross-validation, only the channels that are consistently available throughout the duration of each case are selected for analysis to mitigate the effect of data heterogeneity. In total, 19 channels have been measured for all 13 cases, including Fp1, Fp2, O1, O2, T3, T4, F7, F3, Fz, F4, F8, Cz, C3, T5, P3, Pz, P4, T6, and C4.

2.2. Data pre-processing

The collected data show various artifacts, which can be categorized into physiological and nonphysiological types. Physiological artifacts include influences from electromyography (EMG), electrocardiography (ECG), and electrooculography (EOG), while nonphysiological artifacts include the influence of skin contact and cable movements. To eliminate artifacts in EEG recordings, pre-processing based on Independent Component Analysis (ICA) is carried out following the method proposed in [21]. Based on the common metric for validating the effectiveness of artifact removal, which is the signal-to-noise ratio (SNR), ICA outperformed other artifact removal methods, especially in the case of patient 11. In channels Fp1 and Fp2, the SNR improved significantly after ICA was applied. For example, the SNR for Fp1 increased from -37.2 dB to 14.7 dB, while for Fp2 it increased from -26.8 dB to 15.5 dB. By successfully removing these artifacts, the implementation of ICA allows the acquisition of EEG recordings with improved signal quality. This allows more complex features to be extracted from the EEG signal [21].

3. Compressive sensing

Compressive sensing (CS), also known as compressive sampling or sparse sensing, is a signal processing approach that efficiently compresses signals by reducing the number of samples based on signal sparsity [18]. Compressive sensing allows data to be compressed already during acquisition, which is advantageous for bandwidth-limited systems. It also reduces the complexity and power consumption of the sensor hardware since fewer measurements are needed to reconstruct the original signal [22]. The application of CS is conditioned by the sparsity of the signal [23]. This means that the signal should have a limited non-zero subset of coefficients, either in the time domain or in another transform domain, such as the Fourier transform, the wavelet transform, or the discrete cosine transform (1).

$$X = \psi \times S \quad (1)$$

where ψ is the transform domain matrix and S is the sparse representation of the original signal X . Several studies have shown that the EEG signal does not exhibit sparsity or compressibility in the time

domain [24]. However, their results have indicated that DCT serves as the appropriate transform domain in which the EEG signal exhibits sparsity [25]. In this study, the DCT matrix ψ is used.

Compressive sensing (CS) offers several advantages over traditional compression methods, particularly in the context of EEG signal processing and embedded system applications. Unlike conventional techniques such as JPEG encoding, CS has demonstrated superior performance in preserving signal integrity, measured by metrics such as the structural similarity index (SSIM) [26]. This advantage extends beyond image compression to various signal types, including EEG data [27,28]. One of the key strengths of CS lies in its ability to minimize noise influence and effectively handle missing or corrupted data [29]. This robustness is particularly valuable in EEG signal processing, where artifacts and signal degradation are common challenges. Furthermore, CS can achieve high compression ratios while maintaining signal fidelity, which is crucial for efficient data transmission and storage in resource-constrained environments.

However, CS is not without limitations. The method's effectiveness is contingent upon the signal's sparsity in a particular domain (e.g., time or frequency). This requirement may not always be met for all types of EEG data, potentially limiting its applicability in certain scenarios. Additionally, while the encoding process in CS is relatively simple and can be highly lossy, the reconstruction phase is computationally intensive. This asymmetry in computational requirements makes CS particularly well-suited for embedded systems applications at the acquisition end, where low-power, resource-constrained devices are common.

The computational demand for signal reconstruction in CS requires more advanced computing units, typically a powerful computer or server. This aspect creates a natural division of labor in CS-based systems: low-power embedded devices can efficiently acquire and compress signals, while more capable computing infrastructure handles the complex task of signal reconstruction and subsequent analysis.

3.1. EEG signal compression

Consider $X \in R^{N \times CH}$, where N is the number of samples, and CH denotes the number of channels, this corresponds to the original multichannel EEG signal. $Y \in R^{M \times CH}$ is the compressed version of the original signal where $M \ll N$ represents the measurement number. M should respect the Restricted Isometry Property (RIP) represented in Eq. (2).

$$M = \mathcal{O}\left(k \log\left(\frac{N}{k}\right)\right) \quad (2)$$

where k is the number of the non-zero elements of the signal in the sparse domain [30]. The relation between X and Y in CS is represented by (3).

$$Y = \phi X^T = \phi (\psi S)^T \quad (3)$$

$\phi \in R^{M \times N}$ represents the measurement matrix that should be incoherent with the sparsification matrix ψ to improve the system's ability to capture important information. There are two main measurement matrices used with the EEG signal that are incoherent with the DCT matrix ψ : The Gaussian Random Matrix and the Random Binary Matrix. For a fast and energy-efficient system, the random binary matrix is the appropriate matrix for real-time systems because it is easy to generate and requires less memory than the Gaussian random matrix [25].

To optimize signal recovery in compressed sensing systems, it is crucial to incorporate the sparsification matrix into both the acquisition and reconstruction processes. However, this integration presents significant challenges due to the common separation of acquisition and reconstruction platforms. Typically, data acquisition occurs on resource-constrained devices such as microcontrollers, while reconstruction is performed on more powerful systems like single-board computers (SBCs) or general-purpose computers. This hardware divide is often accompanied by a software divide, with different programming

languages used on each platform. To bridge the gap between the acquisition and reconstruction stages in compressed sensing, we propose the use of a linear congruential generator (LCG) for consistent sparsification (Algorithm 1). LCGs offer deterministic pseudorandom number generation with minimal computational overhead, making them suitable for resource-constrained devices. By initializing LCGs with the same seed and parameters on both the acquisition and reconstruction sides, we guarantee identical sparsification patterns, crucial for accurate signal recovery.

Algorithm 1 Linear Congruential Generator (LCG)

```

1: Constants:  $A = 1664525$ ,  $C = 1013904223$ ,  $M = 2^{32}$ 
2: State:  $seed \leftarrow 1$  ▷ Initial seed
3: function LCG_RANDOM
4:    $seed \leftarrow (A \cdot seed + C) \bmod M$ 
5:   return  $seed$ 
6: end function
7: function PSEUDO_RANDOM_BINARY
8:   return  $(LCG\_RANDOM() \gg 31) \bmod 2$ 
9: end function
10: procedure GENERATERANDOMBITS( $n$ )
11:   for  $i \leftarrow 0$  to  $n - 1$  do
12:     output PSEUDO_RANDOM_BINARY
13:   end for
14: end procedure

```

3.2. EEG signal reconstruction

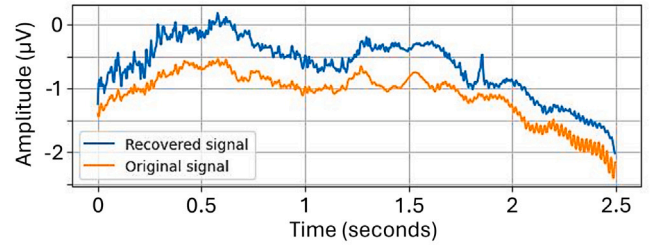
The reconstruction process is a critical phase of CS, with the aim of recovering the original signal X from its compressed measurements Y . Several reconstruction techniques have been developed to enable accurate signal recovery while taking advantage of its intrinsic sparsity. These techniques use mathematical optimizations to solve the under-determined system presented in (4), which aims to estimate the sparse signal S .

$$\min_S \|S\|_1 \text{ s.t. } \phi(\psi S)^T = Y \quad (4)$$

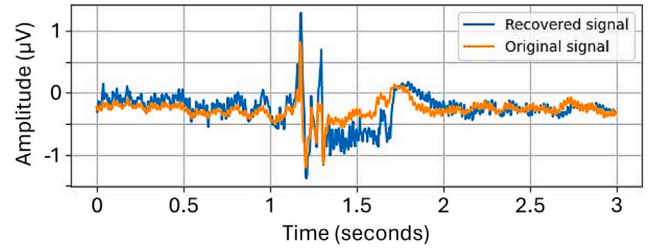
Three categories of reconstruction algorithms have been presented in [31,32], including greedy, Bayesian, and convex optimization methods. Research has shown that the convex optimization family can be computationally intensive for high-dimensional EEG data, while the Bayesian and Greedy algorithms have solved the problem of data dimensionality [33]. In terms of complexity, Bayesian algorithms have a complexity of $\mathcal{O}(N; M^2)$, while Greedy algorithms have a complexity of $\mathcal{O}(N; M)$ [17]. For real-time processing, greedy algorithms, in particular Orthogonal Matching Pursuit (OMP), along with its 2D counterpart Simultaneous OMP (SOMP), prove to be suitable options. In the context of this study, where EEG data is collected from multiple channels, the use of SOMP efficiently and accurately reconstructs the multi-channel EEG signal within a real-time system. To evaluate the effectiveness of the reconstructed EEG signal, the Mean Square Error (MSE) is used [34]. A representation of the MSE equation can be found in (5).

$$MSE = \frac{1}{n} \sum_{i=1}^n (X_i - S_i)^2 \quad (5)$$

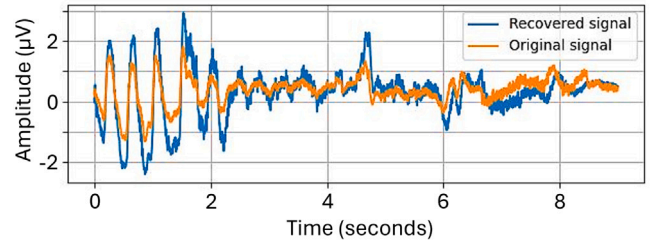
The SOMP algorithm was tested on all files in the epileptic EEG dataset. Fig. 1 shows a compression of 5% between the original and reconstructed EEG signals recorded during different onset episodes (AsTSZ, STSZ, and USZ). This comparison concerns channel 6 of patient 7 (P7-CH6, Fig. 1(a)), channel 17 of patient 5 (P5-CH17, Fig. 1(b)), and channel 8 of patient 11 (P11-CH8, Fig. 1(c)). It is noticeable that the reconstructed signal closely follows the envelope of the original signal, with only minor amplitude distortion. In terms of frequency, the reconstructed signal preserves the inherent characteristics of the original signal, especially in the delta and theta EEG frequency bands.



(a) Original and reconstructed signal for patient 7 (P7) channel 6 (T4) with AsTSZ seizure



(b) Original and reconstructed signal for patient 5 (P5) channel 17 (P4) with STSZ seizure



(c) Original and reconstructed signal for patient 11 (P11) channel 8 (F3) with USZ seizure

Fig. 1. Comparison of original and reconstructed signal for different patients with different seizure types.

MSE serves as our primary metric, providing a direct measure of the average squared difference between the original and reconstructed signals:

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 \quad (6)$$

Where Y_i is the original signal, $\hat{Y}_i$ is the reconstructed signal, and n is the number of samples. Lower MSE values indicate better reconstruction quality. MSE values are presented for three different reconstructed signals, each representing a different type of seizure which are visualized in Fig. 2. Notably, a low MSE value is presented for the different patients, indicating that the reconstructed EEG signal is very similar to the original signal. On the other hand, higher MSE values are observed for certain EEG channels (FP2, P4, T6, and C4), especially for patient 7, indicating a dissimilarity between the two signals, suggesting that some information may have been lost or distorted during the reconstruction process.

To complement MSE, we employ the Pearson correlation coefficient (P_{corr}), which measures the linear correlation between the original and reconstructed signals:

$$P_{corr} = \frac{\sum_{i=1}^n (Y_i - \bar{Y})(\hat{Y}_i - \bar{\hat{Y}})}{\sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2} \sqrt{\sum_{i=1}^n (\hat{Y}_i - \bar{\hat{Y}})^2}} \quad (7)$$

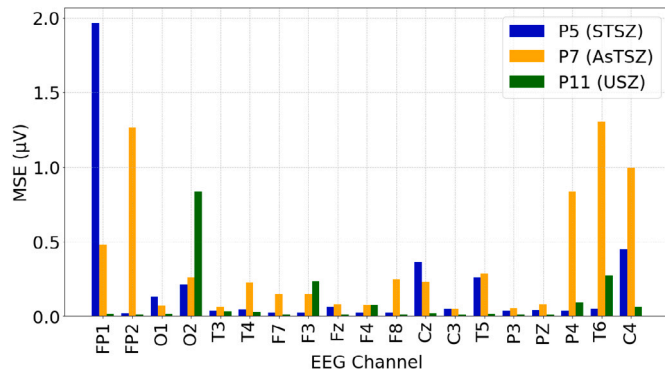


Fig. 2. MSE of the reconstructed EEG channel signals for different patients having different seizure types.

Where $\bar{Y}$ and $\hat{\bar{Y}}$ are the means of the original and reconstructed signals, respectively. P_{corr} ranges from -1 to 1 , with values closer to 1 indicating stronger positive linear relationships. The correlation coefficient provides insights into the preservation of signal patterns and trends, complementing the absolute error information from MSE. A high correlation suggests that the reconstructed signal accurately captures the temporal dynamics of the original EEG, even if the absolute amplitudes might differ.

For a more comprehensive evaluation, we also consider Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM). PSNR is derived from MSE and provides a logarithmic scale of reconstruction quality:

$$PSNR = 10 \log_{10} \left(\frac{MAX_I^2}{MSE} \right) \quad (8)$$

Where MAX_I is the maximum possible signal value.

SSIM, adapted from image processing [35], assesses structural similarity:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (9)$$

Where x and y are the original and reconstructed signals, μ and σ represent their respective means and variances, and c_1, c_2 are stabilizing constants.

This combination of metrics, with MSE as the primary measure, provides a nuanced understanding of CS reconstruction quality. It balances absolute error (MSE, PSNR) with assessments of linear relationship strength (P_{corr}) and structural preservation (SSIM), offering a comprehensive evaluation of the reconstructed EEG signals.

3.3. Selection of optimal compression ratio

To determine the optimal compression ratio (CR) for the reconstruction of CS-based EEG signals, we performed a comprehensive analysis across various CRs, as shown in Table 2. The analysis considered multiple factors, including PSNR, SSIM, Pearson correlation coefficient (P_{corr}), transmission time, and energy consumption.

The signal quality metrics show remarkable consistency across a wide range of compression ratios. For instance, the Peak Signal-to-Noise Ratio (PSNR) values for 5% (16.33 ± 0) and 70% (16.219 ± 0.013) compression ratios are notably close, indicating that signal fidelity is well-preserved even at higher compression levels. Similarly, the Pearson correlation coefficients (P_{corr}) for both ratios (0.74 ± 0 at 5% and 0.734 ± 0.001 at 70%) show minimal difference, suggesting comparable signal integrity. While there is a more noticeable difference in Structural Similarity Index (SSIM) values (0.526 ± 0.001 at 5% vs 0.397 ± 0.006 at 70%), both ratios maintain reasonable structural information preservation.

The most significant differences are observed in transmission time and energy consumption. However, these improvements are gradual across the range of compression ratios, rather than showing a stark contrast between low and high compression. Transmission time decreases from 20.274 ± 0.003 s at 5% CR to 6.431 ± 0.002 s at 70% CR, while energy consumption shows a modest reduction from 12.5 ± 2.121 mWh at 5% CR to 4.5 ± 0.707 mWh at 70% CR. The reduction in data size is also gradual, moving from 187.1 kB at 5% CR to 125.35 kB at 70% CR.

Given these observations, it appears that the EEG signal maintains its essential characteristics across a wide range of compression ratios. The choice between lower and higher compression ratios may depend more on specific application requirements, such as storage constraints or transmission bandwidth, rather than on significant differences in signal quality. For applications where transmission speed and energy efficiency are critical, higher compression ratios like 70% offer noticeable benefits without substantially compromising signal quality. However, for scenarios where minor improvements in signal fidelity are valued over efficiency gains, lower compression ratios like 5% remain viable options.

3.4. Analysis of compressed sensing evaluation

Table 3 presents a comprehensive evaluation of Compressive Sensing (CS) applied to EEG data from 13 patients, using a 70% compression ratio. The results demonstrate significant improvements in data efficiency, with consistent file size reductions of 70% across all patients. This compression translates to substantial decreases in transmission times, as exemplified by Patient 9, where the time was reduced from 783.559 to 121.580 s. Overall, the total file size decreased from 15029 kB to 4509.54 kB, while the cumulative transmission time saw an 84% reduction, from 2518.532s to 400.392s.

However, the impact on signal quality metrics reveals a more nuanced picture. The Peak Signal-to-Noise Ratio (PSNR) exhibits considerable variability, ranging from 7.946 to 23.376, with a mean of 16.15 ± 3.98 . Similarly, the Structural Similarity Index (SSI) spans from 0.118 to 0.657, averaging 0.34 ± 0.18 , while the Pearson Correlation Coefficient (P_{corr}) ranges from 0.415 to 0.884, with a mean of 0.68 ± 0.15 . This variability suggests that the 70% compression ratio, while efficient, may not uniformly preserve signal integrity across all patients.

Given these observations, it becomes apparent that the 70% compression ratio, despite its impressive efficiency gains, may not be suitable for all scenarios in EEG data analysis. The heterogeneity in signal quality metrics across patients indicates potential compromises in data integrity for some individuals. In light of these considerations, employing a more conservative 5% compression ratio could be preferable in certain situations, particularly where maintaining high fidelity of EEG patterns is crucial. While a 5% ratio will not match the size reduction and transmission time improvements of 70% compression, it offers a more balanced approach. This is particularly valuable in applications prioritizing data integrity, such as clinical settings or research phases where the impact of compression on analytical outcomes is not fully understood. Given these considerations, the remainder of this manuscript will use a 5% compression ratio as a proof of concept. This choice aims to demonstrate the feasibility of compressive sensing in EEG analysis while prioritizing data fidelity and robustness of results.

4. Feature extraction and selection

Differentiating between the epileptic EEG patterns of compressed AsTSZ, STSZ, and USZ is a challenging task due to the variability observed in these patterns. This variability depends on factors such as the location and intensity of the seizure [36]. Compressed raw EEG signals exhibit non-linear dynamic properties, making it difficult for classifiers to perform well in epilepsy classification. To improve

Table 2
Compressive sensing evaluation under different compression ratios (Example: Patient 12)

Compression Ratio (CR)	Size (kB)	PSNR	SSIM	P_{corr}	Transmission Time (s)	Energy (mWh)
0	191.85	16.336 ± 0	0.530 ± 0.000	0.741 ± 0.000	21.685 ± 0.000	11.5 ± 0.707
5	187.10	16.330 ± 0	0.526 ± 0.001	0.740 ± 0.000	20.274 ± 0.003	12.5 ± 2.121
10	182.35	16.332 ± 0	0.267 ± 0.359	0.413 ± 0.465	19.505 ± 0.008	8.5 ± 0.707
15	177.60	16.322 ± 0.004	0.507 ± 0.000	0.740 ± 0.000	18.422 ± 0.003	8.0 ± 1.414
20	172.85	16.316 ± 0.007	0.501 ± 0.002	0.740 ± 0.001	17.021 ± 0.004	7.5 ± 0.707
25	168.10	16.327 ± 0.016	0.500 ± 0.005	0.741 ± 0.001	16.248 ± 0.004	6.5 ± 0.707
30	163.35	16.339 ± 0.002	0.490 ± 0.004	0.741 ± 0.000	15.161 ± 0.002	7.0 ± 0.000
35	158.60	16.321 ± 0.001	0.473 ± 0.011	0.740 ± 0.000	13.851 ± 0.004	6.0 ± 0.000
40	153.85	16.288 ± 0.021	0.452 ± 0.001	0.738 ± 0.001	12.985 ± 0.001	6.0 ± 0.000
45	149.10	16.278 ± 0.044	0.445 ± 0.012	0.737 ± 0.003	11.711 ± 0.001	6.0 ± 0.000
50	144.35	16.284 ± 0.004	0.425 ± 0.003	0.738 ± 0.001	10.807 ± 0.007	5.5 ± 0.707
55	139.60	16.252 ± 0.028	0.413 ± 0.013	0.736 ± 0.002	9.725 ± 0.000	5.5 ± 0.707
60	134.85	16.270 ± 0.030	0.423 ± 0.013	0.737 ± 0.002	8.445 ± 0.000	5.5 ± 0.707
65	130.10	16.217 ± 0.000	0.214 ± 0.275	0.459 ± 0.390	7.512 ± 0.001	5.5 ± 0.707
70	125.35	16.219 ± 0.013	0.397 ± 0.006	0.734 ± 0.001	6.431 ± 0.002	4.5 ± 0.707
75	120.60	16.213 ± 0.034	0.394 ± 0.003	0.734 ± 0.002	5.347 ± 0.002	4.5 ± 0.707
80	115.85	16.143 ± 0.000	0.384 ± 0.000	0.729 ± 0.000	4.191 ± 0.000	4.0 ± 0.000
85	111.10	16.209 ± 0.114	0.353 ± 0.017	0.733 ± 0.007	3.186 ± 0.001	3.5 ± 0.707
90	106.35	16.046 ± 0.006	0.295 ± 0.021	0.723 ± 0.000	2.064 ± 0.001	3.0 ± 0.000
95	101.60	15.847 ± 0.438	0.182 ± 0.036	0.709 ± 0.030	0.999 ± 0.003	2.0 ± 0.000

Table 3
Compressive sensing evaluation under a 70% compression ratio.

Patient	File Size Before Compression (kB)	File Size After Compression (kB)	PSNR (dB)	SSI	Pcorr	Uncompressed Data Transmission Time (s)	Compressed Data Transmission Time (s)
1	494.00	148.29	18.620	0.323	0.738	37.21	12.178
2	266.00	79.86	19.074	0.657	0.884	45.69	7.098
3	342.00	102.72	19.265	0.599	0.853	58.66	9.128
4	1824.00	547.29	23.376	0.592	0.769	313.527	48.64
5	2090.00	627.00	12.494	0.390	0.760	356.855	55.409
6	494.00	148.29	13.970	0.118	0.617	84.840	13.178
7	532.00	159.72	13.501	0.202	0.567	91.482	14.192
8	1520.00	456.00	13.174	0.163	0.508	261.153	40.525
9	4522.00	1356.72	7.946	0.135	0.415	783.559	121.580
10	2318.00	695.43	15.038	0.205	0.488	398.275	61.806
11	418.00	125.43	18.970	0.291	0.721	71.409	11.082
12	95.00	28.50	16.586	0.381	0.766	6.306	2.530
13	114.00	34.29	17.924	0.323	0.818	9.566	3.046
Mean	–	–	16.15 ± 3.98	0.34 ± 0.18	0.68 ± 0.15	–	–
Sum	15 029	4509.54 (–70%)	–	–	–	2518.532	400.392 (–84%)

classifier efficiency, specific EEG patterns have been used, including low voltage, fast activity, rhythmic theta discharge, complex wave, ictal epileptiform discharges, and polyspike waves [37]. Researchers have used various EEG features, such as spikes, sharp waves, or spike-and-wave complexes, to detect epileptiform activity in EEG signals based on time–frequency analysis [38]. These features help extract relevant information from EEG signals, which is then used by classifiers to categorize different EEG signals [39]. The features of this group are generally used to detect the evolution or fluctuation of EEG patterns during seizure onset [40]. Four features are selected to classify AsTSZ, STSZ, and USZ, including kurtosis (KURT), zero-crossing rate (ZCR), energy (EN), and power [41]. The mathematical expression of each feature is defined as in (10), (11), (12), and (13), respectively [42].

$$Kurt[X] = E\left[\left(\frac{X - \mu}{\sigma}\right)^4\right] = \frac{E[(X - \mu)^4]}{(E[(X - \mu)^2])^2} = \frac{\mu_4}{\sigma^4} \quad (10)$$

Where μ is the mean of the distribution X , and σ is the standard deviation.

$$ZCR = \frac{1}{2W_L} \sum_{i=1}^{W_L} |\text{sgn}[X_i(n)] - \text{sgn}[X_i(n-1)]| \quad (11)$$

$$\text{sgn}(S) = \begin{cases} 1, & \text{if } S \geq 0 \\ -1, & \text{if } S < 0 \end{cases}$$

Where W_L is the length of a signal X and sgn is the sign function.

$$\text{Energy} = \sum_{i=1}^N (cD)^2 \quad (12)$$

Where cD are the N details coefficients.

$$P_x = \sum_{k=1}^N \frac{|X[k]|^2}{S_2 f_s} f_{res} \quad (13)$$

Where f_s is the sampling frequency, $f_{res} = f_s/N$, $S_2 = \sum_{j=1}^N |W[j]|^2$, $w[j]$: is the j 'th DWT window coefficient, and N is the number of samples per window.

When analyzing EEG signals in the time domain, a high kurtosis value indicates a localized and peaked distribution, which may indicate the presence of certain EEG patterns associated with epileptic events, such as spikes or sharp-wave discharges [43]. On the other hand, a lower kurtosis value indicates a flatter distribution, which may be associated with more dispersed or generalized activity [44]. The zero crossing rate reflects the frequency of oscillatory activity and can detect changes in signal polarity or the appearance of rapid transitions [45]. In frequency domain analysis, the energy distribution examines the energy across different frequency components to gain insight into signal characteristics. The wavelet energy function divides the EEG signal into frequency bands, including delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), and beta (12–30 Hz) [46]. The maximum power per band (MPPB) shows the major frequency components and their respective contributions. This helps to identify frequency patterns associated with epileptic seizures, including high-frequency oscillations (HFOs) or irregularities in power distribution across different frequency bands [47].

In this study, multichannel EEG signals from 13 patients were analyzed. Each EEG channel was analyzed separately and relevant features

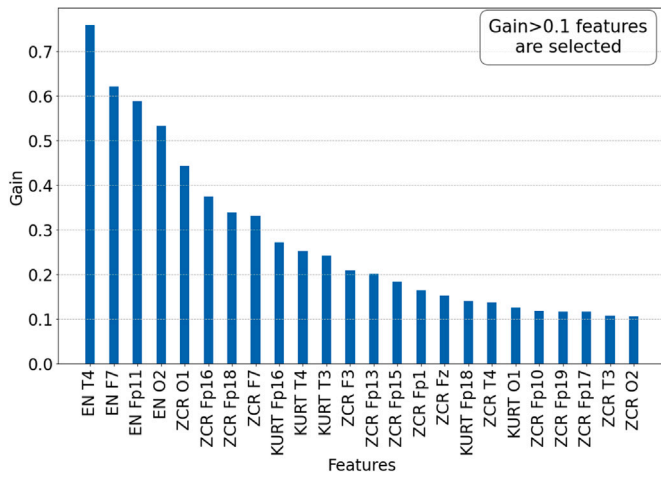


Fig. 3. Gain of the selected 24 EEG features for all EEG recordings.

Table 4
Selected features.

Feature	Selected channel
Zero-crossing rate (ZCR)	Ch1 (Fp1), Ch3 (O1), Ch4 (O2), Ch 5 (T3), Ch6 (T4), Ch7 (F7), Ch8 (F3), Ch9 (Fz), Ch10 (F4), Ch13 (C3), Ch15 (P3), Ch17 (P4), Ch19 (C4)
Kurtosis (KURT)	Ch3 (O1), Ch5 (T3), Ch6 (T4), Ch16 (Pz), Ch18 (T6)
Energy (EN)	Ch4 (O2)

were extracted from 0.5-second overlapping windows to classify AsTSZ, STSZ, and USZ. To overcome the challenge of processing numerous extracted features per EEG channel, the researchers minimized the dimensionality of the feature set and focused on significant elements for accurate seizure classification [48]. To this end, the feature selection method is used to increase computational efficiency, prevent overfitting [49], and reduce classifier complexity [1,50]. The mutual information (MI) method is used to quantify the dependency between features and target variables in the EEG dataset. In the first step, a random threshold of 0.1 was set for the MI gain. This successfully reduced the number of features from 152 to a more manageable 24 features. The feature gain is shown in Fig. 3, where the features are arranged in descending order based on their gain. Upon examination of the selected features, it was found that the feature associated with MPPB does not significantly influence the classification. This is due to the lack of distinct patterns or variations in power distribution across frequency bands that can differentiate between AsTSZ, STSZ, and USZ.

As an integral step in the feature selection process, it is critical to assess the correlation between features within all EEG channels. This allows for the identification and subsequent elimination of features that have significant intra-channel correlations. For this purpose, the correlation matrix shown in Fig. 4 is used to quantify the dependency between features and target variables in the EEG dataset. Based on the analysis of the correlation matrix, it was determined that five additional features, namely 'EN CH6', 'EN CH7', 'EN CH11', 'ZCR CH16', and 'ZCR CH18', had a high degree of correlation coefficient with other variables, exceeding the predetermined threshold of 0.9. Therefore, to reduce multicollinearity problems, these features were deliberately excluded from the analysis. As a result, a final set of 19 features was obtained to classify AsTSZ, STSZ, and USZ. These are summarized in Table 4.

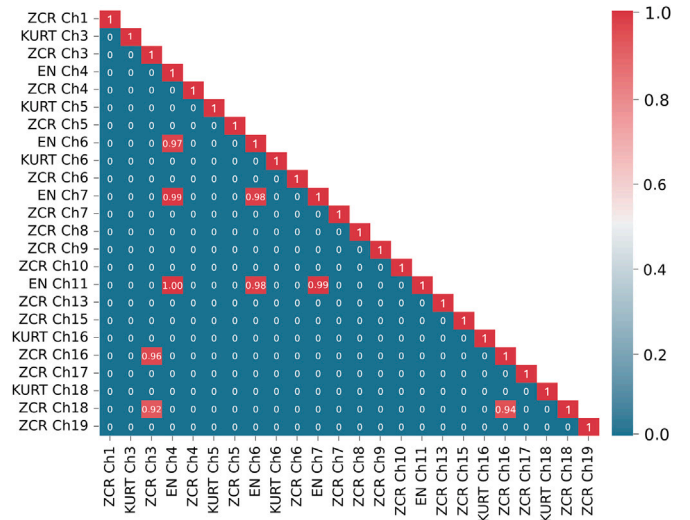


Fig. 4. Correlation matrix of the 24 selected features for reducing the number of features to 19 for all EEG dataset and for every type of seizure.

5. Hardware setup and model implementation

5.1. Hardware setup

The proposed compressive sensing (CS) method for EEG signal processing is implemented using a dual-platform system, as illustrated in Fig. 5. The figure depicts a comprehensive setup, comprising microcontroller boards and a power measurement device. The system employs the complementary capabilities of two distinct hardware platforms: an STM32 microcontroller for data acquisition and compression and a Raspberry Pi 4 Model B for signal reconstruction and analysis.

In contrast with direct EEG acquisition, the system simulates the transmission of EEG data. As illustrated in the accompanying image, the Raspberry Pi 4 Model B, delineated in blue, serves as the data source. The system stores EEG signals as EDF (European Data Format) files and transmits them via UART to the STM32 microcontroller. The data is transmitted in the form of double-precision floating-point values in a little-endian format. To ensure the stability and reliability of UART communication and prevent the reception buffer from overflowing, a 50 ms delay is incorporated between successive transmissions. Additionally, Direct Memory Access (DMA) is utilized to further minimize data corruption and optimize data transfer efficiency.

The STM32 microcontroller, indicated in the illustration with a green outline, serves as the compression unit. The CS algorithm is then applied for data compression, with the simulated EEG data received from the Raspberry Pi being the source. For signals exceeding approximately 10 s (2800 samples), the system employs a paging mechanism, whereby data is compressed in segments of 2800 values, thus enabling efficient management of memory constraints. Subsequently, the STM32 transmits the compressed data back to the Raspberry Pi via UART for reconstruction and analysis.

The system's energy efficiency represents a critical consideration in its design. To accurately measure power consumption, a KOWSI USB power meter, indicated by an orange outline in the image, is integrated into the setup. This device provides real-time energy and time measurements, enabling comprehensive analysis of the system's performance and efficiency.

5.2. Model selection and evaluation

Machine learning has been shown to be effective in interpreting EEG signals for various diagnostic purposes of epilepsy, such as the detection

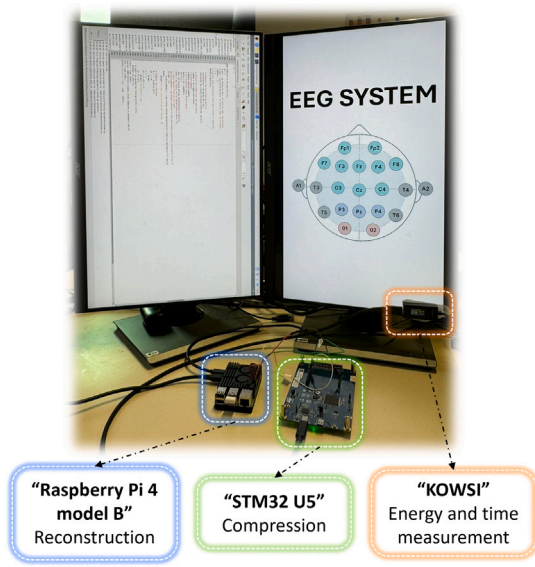


Fig. 5. Proposed system design for seizure classification based on CS.

of normal and abnormal epileptic signals and classification of seizure status (preictal, ictal, and postictal) [25], and for other classification purposes [51]. The possibility of a machine learning approach that learns to interpret the shape of EEG signals to classify epileptic seizure types appears to be a feasible solution [14]. Machine learning is often used as a suitable approach for signal processing. Random Forest (RF), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Extreme Gradient Boosting (XGBoost) are the most commonly used classifiers.

The goal of this study is to develop and implement a two-stage approach to classify AsTSZ, STSZ, and USZ seizures with fewer measurements using compressive sensing. The initial phase focuses on evaluating the feasibility of these classifiers to effectively discriminate between seizure types using features extracted from the reconstructed EEG dataset. The classifiers are then evaluated for their performance using relevant evaluation metrics, including accuracy, precision, recall, and f1 score. In the next step, the most promising algorithm identified based on its performance will be implemented in hardware to classify AsTSZ, STSZ, and USZ seizures using features extracted from the compressed EEG dataset to facilitate real-world testing and practical application.

The performance of the developed machine learning model depends on various hyperparameters, which can be set manually before explicitly training the model. The hyperparameters used for the SVM, RF, KNN, and XGBoost models are detailed in Table 5. In this study, the data manipulation is done using the Python tool. The data set is divided into 80% for training, 20% for testing, and 5 k-fold validation. Upon training the models developed based on the optimal hyperparameters from Table 5 and conducting 10 consecutive tests, the SVM and kNN models both attained an average classification accuracy of 88.40%, with a precision of 88.49%, a recall of 88.60%, and an f1 score of 88.38%. In contrast, the RF model achieved a higher average accuracy of 94.17%, surpassing the SVM and kNN models in classifying AsTSZ, STSZ, and USZ. However, when considering the four models tested, the XGBoost model emerged as the top performer. XGBoost showed an impressive average classification accuracy of 98.73%, accompanied by a precision of 98.78%, a recall of 98.73%, and an f1-score of 98.72%. Detailed results for all models developed can be found in Table 6.

At this stage, it is clear that the XGBoost model performs well in classifying the three types of seizures using features extracted from the original EEG data. The next challenge is to demonstrate the practicality

Table 5

Selected hyper-parameters for classification models.

Predictive model	Hyperparameter	Tuned to
SVM	kernel	linear
	gamma	scale
RF	n_estimators	100
	max_depth	2
	samples_split	1
	samples_leaf	none
	bootstrap	false
KNN	weights	distance
	K-neighbors	7
	metric	manhattan
XGBoost	n_estimators	1000
	max_depth	8
	learning_rate	0.3
	subsample	0.6
	colsample_bytree	0.8
	gamma	0.2

Table 6

Machine learning models evaluation.

Predictive model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
SVM	88.60	88.49	88.60	88.38
KNN	88.60	88.49	88.60	88.38
RF	94.17	94.29	94.17	94.19
XGBoost	98.73	98.78	98.73	98.72

and applicability of this solution in real-world situations by implementing and evaluating the XGBoost model in hardware using features extracted from the compressed EEG signal, exploiting the fusion of machine learning and compressive sensing.

The XGBoost model architecture for seizure type classification from EEG data, is shown in Fig. 6. It employs a structured approach for efficient data processing. It utilizes Gradient Boosting to iteratively construct decision trees, identifying seizure-specific patterns in the EEG data. To prevent overfitting and reduce complexity, the model incorporates a regularized objective function and tree pruning. During training, decision trees are built and adjusted based on the input data. A decision point determines whether hyperparameter tuning is needed, optimizing parameters such as learning rate, maximum depth, and sub-sample ratio. Once optimized, the model generates predictions classifying seizure types. This architecture strikes a balance between sophisticated analysis and measures to ensure accuracy and generalizability in seizure classification.

The next step is to implement this model in hardware using features extracted from compressed EEG signals to demonstrate its practicality and applicability in real-world seizure monitoring systems.

5.3. Model implementation and evaluation

The implementation phase involves developing a seizure classification system comprising transmitter and receiver nodes, as illustrated in Fig. 7. This figure presents a comprehensive workflow divided into four main stages: data acquisition and filtering, compressive sensing, feature extraction and selection, and machine learning model implementation and evaluation.

In the first stage, EEG data is acquired from 19 channels and filtered using high-pass (10 Hz) and low-pass (100 Hz) filters to remove noise and isolate relevant frequency bands. The second stage implements compressive sensing (CS) at the transmitter node, which is performed on an STM32 microcontroller. CS involves applying a DCT transform followed by multiplication with a random matrix, resulting in a compressed signal suitable for wireless transmission. The receiver, implemented on a Raspberry Pi, performs inverse DCT and EEG sparse reconstruction using SOMP algorithm. The third stage involves feature

Table 7
Summary of CS-based EEG studies for seizure detection and classification.

Studies	Application	Sensing Matrix	Reconstruction Algorithms	Sparse Basis	Classifier	Evaluation Matrices	Hardware Implementation
[52]	Seizure detection	Random	Reconstruction free	Gabor, DCT	SVM	CR, execution time	$\times$
[53]	Seizure detection	Random	Norm-2	–	SVM	CR, Accuracy	$\times$
This work	Seizure classification	Random Binary	SOMP	DCT	XGBoost	CR, Memory, PSNR, SSI, MSE, P_corr, Time, Energy, Accuracy (98.73% for 5% CR)	✓

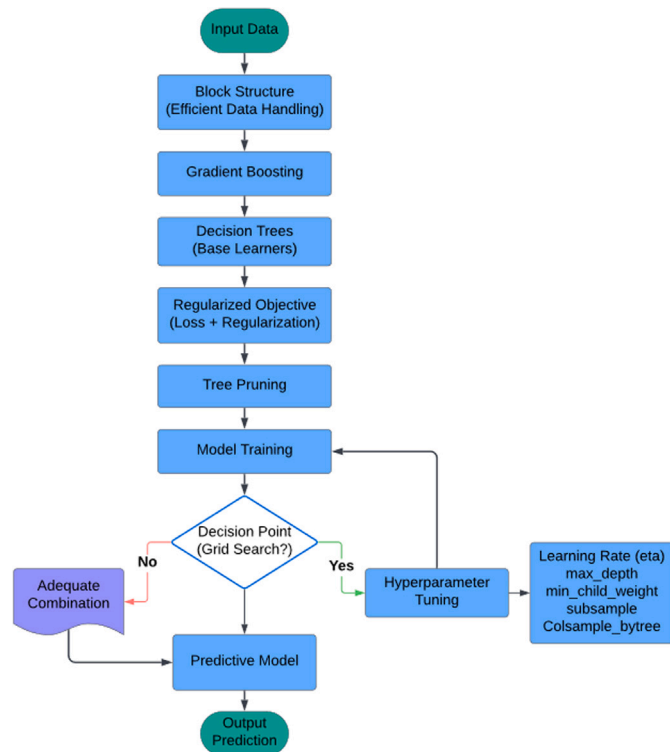


Fig. 6. XGBoost architecture for seizures classification.

extraction and selection, where features with gain >0.1 are selected, emphasizing low complexity, fast processing, and high precision. The final stage involves machine learning model implementation and evaluation, utilizing a robust framework of training, validation, and testing sets, with hyperparameter tuning and model validation.

During the model training phase, system evaluation was conducted on a Dell PC equipped with an Intel(R) Core(TM) i5-1035G1 CPU (base: 1.00 GHz, turbo: 3.60 GHz) and 8.00 GB RAM, running a 64-bit OS with an x64 architecture based on Python Jupyter software. The same python code is later used in Raspberry Pi 4.

6. Comparison with state-of-the-art

Recent advancements in EEG signal classification for seizure detection have shown promising results. [10] applied blind source separation (BSS) and Linear Discriminant Analysis (LDA), achieving 80.72% accuracy. [11] combined deep RNN and CNN with ICA and CCA, reaching 82.00% accuracy. [12] utilized Tunable-Q Wavelet Transform (TQWT), entropies, and Particle Swarm Optimization (PSO) with ANN, attaining 88.80% accuracy. Notably, [14] implemented a hardware-based solution using FPGA technology and a feed-forward MLP ANN, achieving the highest accuracy of 95.14%. These studies demonstrate significant progress in seizure classification, ranging from traditional

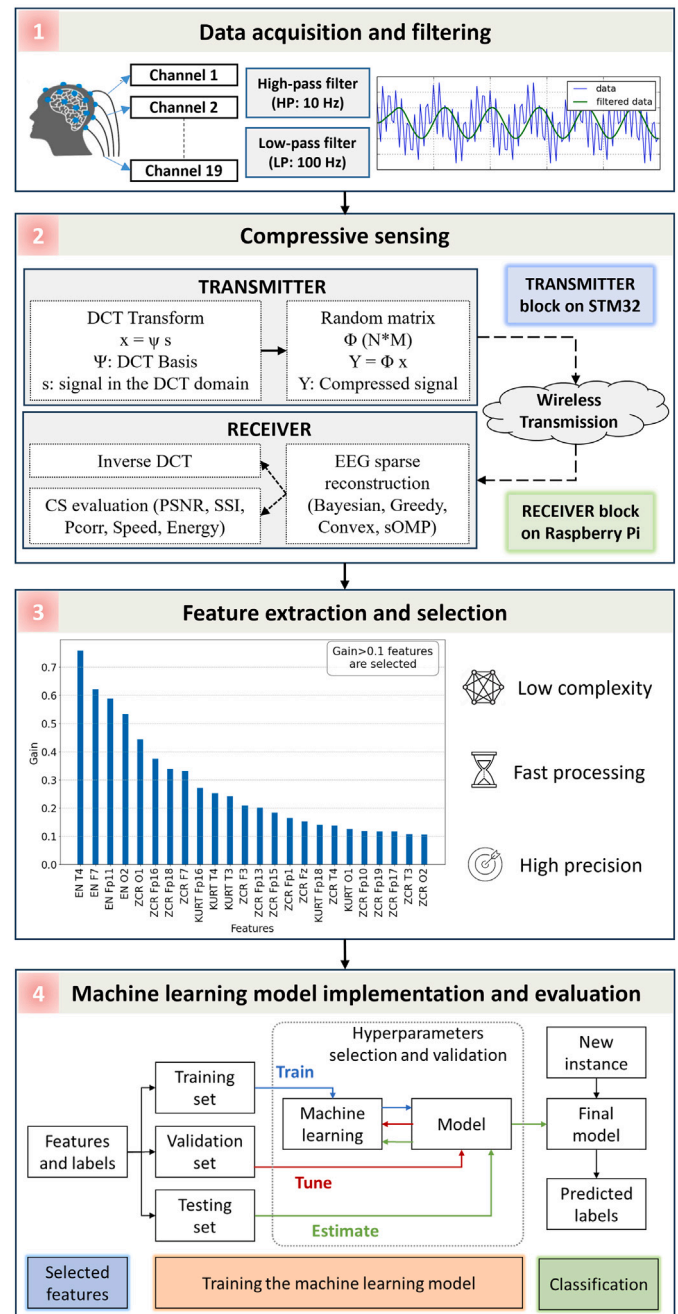


Fig. 7. Workflow of the proposed work.

machine learning to advanced deep learning and hardware implementations, with the FPGA-based approach showing particular promise for real-time applications.

Recent studies have explored compressive sensing (CS) techniques for EEG-based seizure detection and classification. Table 7 summarizes key CS-based EEG studies. [52] applied random sensing with reconstruction-free Gabor and DCT sparse basis, using SVM for classification, evaluated by CR and execution time. [53] also utilized random sensing, Norm-2 reconstruction, SVM classifier, and CR and accuracy as evaluation metrics. However, these studies lack hardware evaluation and only focus on detecting seizures rather than classifying them. Our work introduces a novel approach using random binary sensing matrix, SOMP for reconstruction, DCT as sparse basis, and XGBoost for classification. We evaluate using comprehensive metrics including CR, MSE, F-score, time, energy, memory, and PSNR, achieving 98.73% accuracy. Notably, our method is the only one among these studies to implement hardware acceleration, potentially offering real-time processing capabilities for seizure detection systems.

7. Conclusion

This study has demonstrated the potential of machine learning-based systems, particularly the fusion of machine learning and compressive sensing, to address the urgent issue of real-time epilepsy diagnosis. The primary objective was to develop an automatic classification system that would be characterized by high accuracy while also considering critical factors such as complexity, speed, and response time. By employing the compressive sensing approach in an innovative manner, this research has successfully demonstrated the creation of an efficient seizure classification system. Using the compressive sensing (CS) approach for feature extraction and an XGBoost classifier, we achieved a notable classification accuracy of 98.78%. The results demonstrate the efficacy of high compression ratios while maintaining a high-quality reconstruction of the signals.

At a 70% compression ratio, a significant reduction in file size (70% average decrease) and transmission time (84% reduction) was observed, with minimal impact on signal quality (PSNR: 16.15 ± 3.98 dB, P_{corr} : 0.68 ± 0.15). The energy efficiency gains were similarly noteworthy, with energy consumption decreasing from 11.5 ± 0.707 mWh at 0% compression to 4.5 ± 0.707 mWh at 70% compression for Patient 12. These findings highlight the potential of our approach in developing portable epilepsy diagnostic systems that offer significant data reduction, enhanced energy efficiency, and reduced transmission times while maintaining classification accuracy.

This study serves to demonstrate the technical feasibility of the approach, paving the way for clinical trials. Further studies are required to expand the dataset with additional epileptic seizures and patient demographics. The objective is to develop a dynamic compression ratio algorithm that adapts to individual patient characteristics, optimizing signal quality across heterogeneous epileptic populations. Additionally, the integration of advanced machine learning techniques will be explored to enhance the adaptive capabilities of the system. By addressing these limitations and pursuing these research directions, the aim is to improve the robustness, versatility, and clinical applicability of the method in epilepsy monitoring and diagnosis, potentially revolutionizing the field of portable epilepsy diagnostic systems.

CRedit authorship contribution statement

Achraf Djemal: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Ahmed Yahia Kallel:** Writing – review & editing, Software. **Cherif Ouni:** Software. **Rihem El Baccouch:** Software. **Dhouha Bouchaala:** Writing – review & editing. **Fatma Kammoun Feki:** Resources. **Chahnez Charfi Triki:** Resources. **Ahmed Fakhfakh:** Supervision, Project administration, Funding acquisition. **Olfa Kanoun:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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